

If the aircraft has been cold soaked at temperatures below -18°C and a battery start must be made, preheat the engine fuel control area and battery if equipment is available and conditions allow.

If the aircraft has been cold soaked and a battery start must be made without preheating the battery, remove and store the battery until it is required if conditions allow. Store the battery in an area where it can be maintained or warmed to a temperature above ambient outside conditions or to approx 20°C .

If stagnated starts are encountered, maintenance action is required.

Due to variations in jet fuels available for commercial operation during cold weather starting, the engines may experience a short delay before ignition after each power lever is moved to just before the IDLE position. This delay should be less than 3 seconds regardless of the type of fuel used. If the ignition delay exceeds 3 seconds, return the power lever to OFF and ventilate the engine for 30 seconds to remove excessive fuel from the combustion section.

In some instances N_1 may accelerate slowly through the 25-30% speed range on a battery start after an engine has been cold soaked and not preheated. If the start is not completed within the starter engagement time limits, shut down the engine. Before attempting the next start, wait for at least one minute to pass. This wait will allow residual heat from the previous start attempt to soak back into the engine and battery and improve conditions for the next start attempt.

At temperatures below -30°C OAT, initial oil pressure indication may be delayed for up to 10 seconds on each engine. The final pressure build-up indication of the LH engine may be delayed for up to 50 seconds.

Depending upon the OAT, a ground idle time of up to 20 minutes may be necessary to bring engine and transmission oils up to operating temperatures.

4.11 FLIGHT CHARACTERISTICS

4.11.1 Flight Controls

CAUTION AVOID EXTREME CYCLIC STICK DISPLACEMENTS WHEN ON THE GROUND WITH ROTOR TURNING.

During ground operations with rotor turning, the cyclic stick must remain in the neutral position and the collective pitch lever in the full down position; however, for functional test purposes, minimum control movements (not more than 3 cm from neutral) are allowed.

Avoid extreme pedal movements during ground operations.

4.11.2 Lateral Control Characteristics

WARNING AVOID STEEP RIGHT TURNS AT AIRSPEEDS BELOW 70 KIAS CLOSE TO THE GROUND IN ORDER TO MAINTAIN SUFFICIENT LATERAL CONTROL MARGIN FOR RECOVERY.

Lateral control characteristics of single-rotor helicopters are significantly influenced by forward speed (airspeed), bank angles and relating g-loads.

During high load factor maneuvers at airspeeds below 70 KIAS the cyclic stick can reach the left lateral stop before retreating-blade stall entry.

Reduction of the load factor and/or nose-left side slip will improve left lateral control margin.

The recommended maximum bank angles for left and right turns at various airspeeds are as follows:

AIRSPEED (KIAS)	MAX ANGLE OF BANK
0 - 30	30°
30 - 60	45°
60 - ($V_{NE}^* - 30$)	60°
($V_{NE}^* - 30$) - V_{NE}^*	30°

* For V_{NE} values, refer to the V_{NE} table in Section 2.